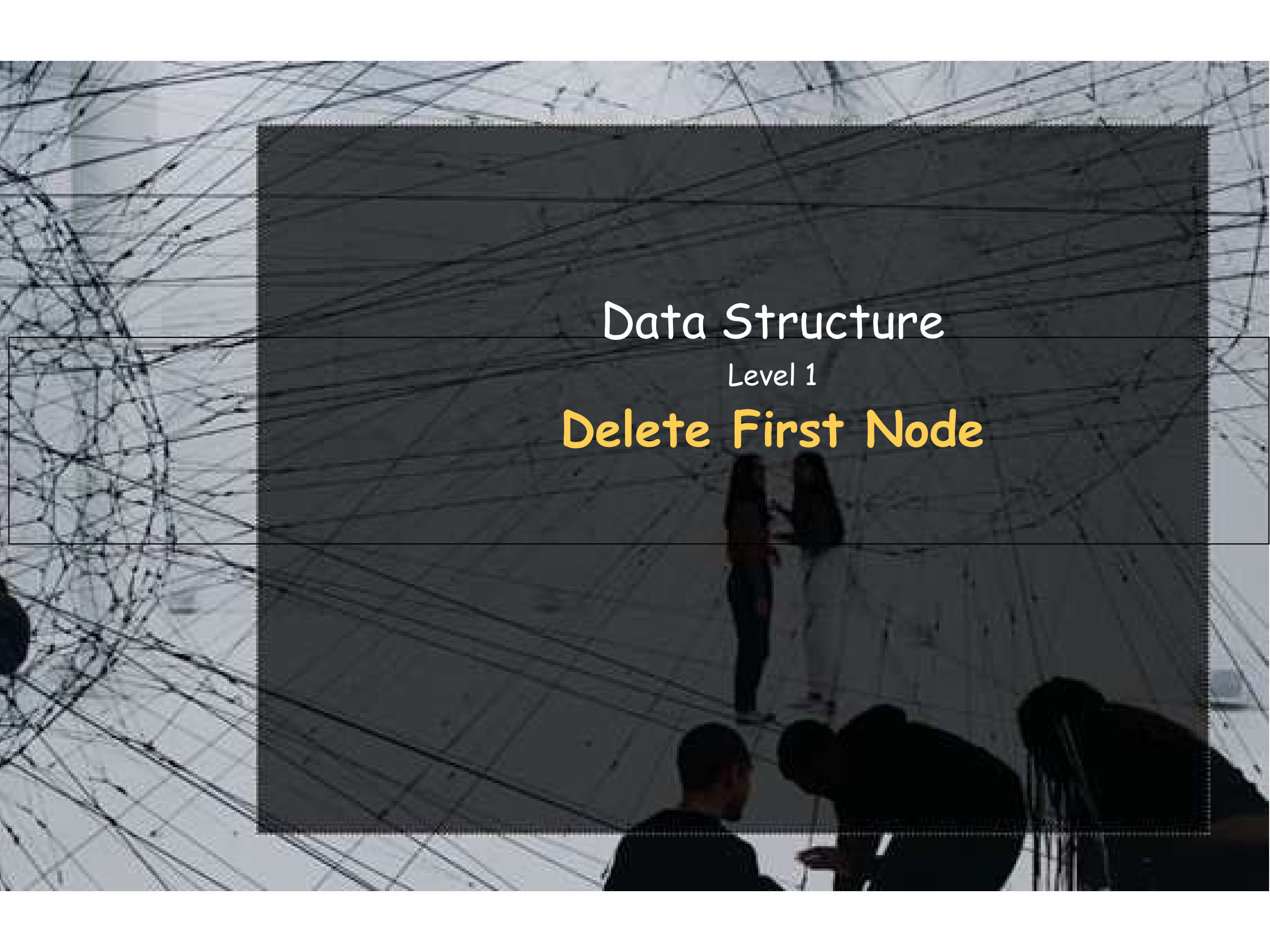


Data Structure

Level 1

Doubly Linked List Operation

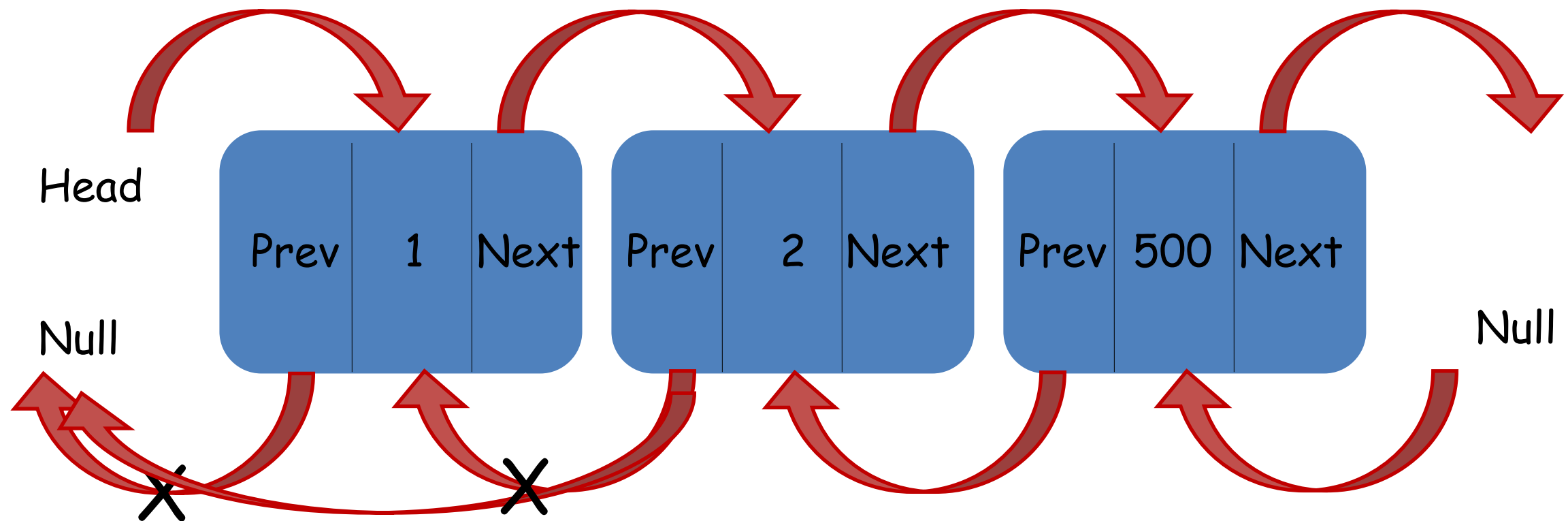


Data Structure

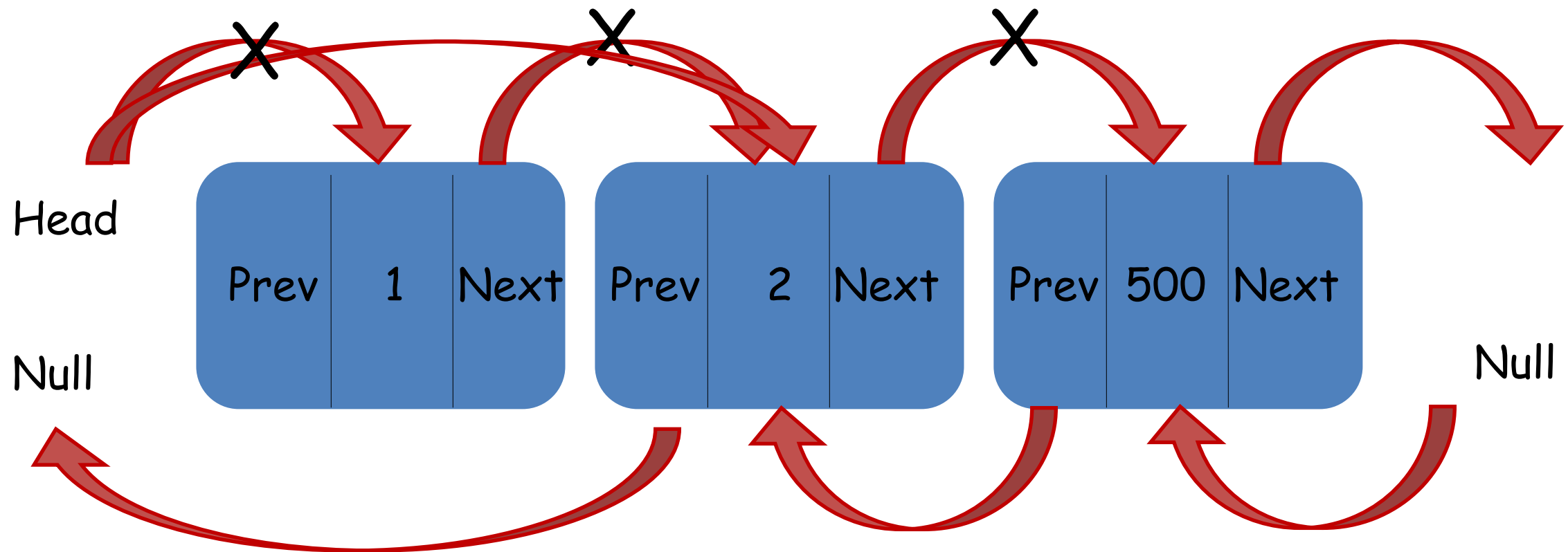
Level 1

Delete First Node

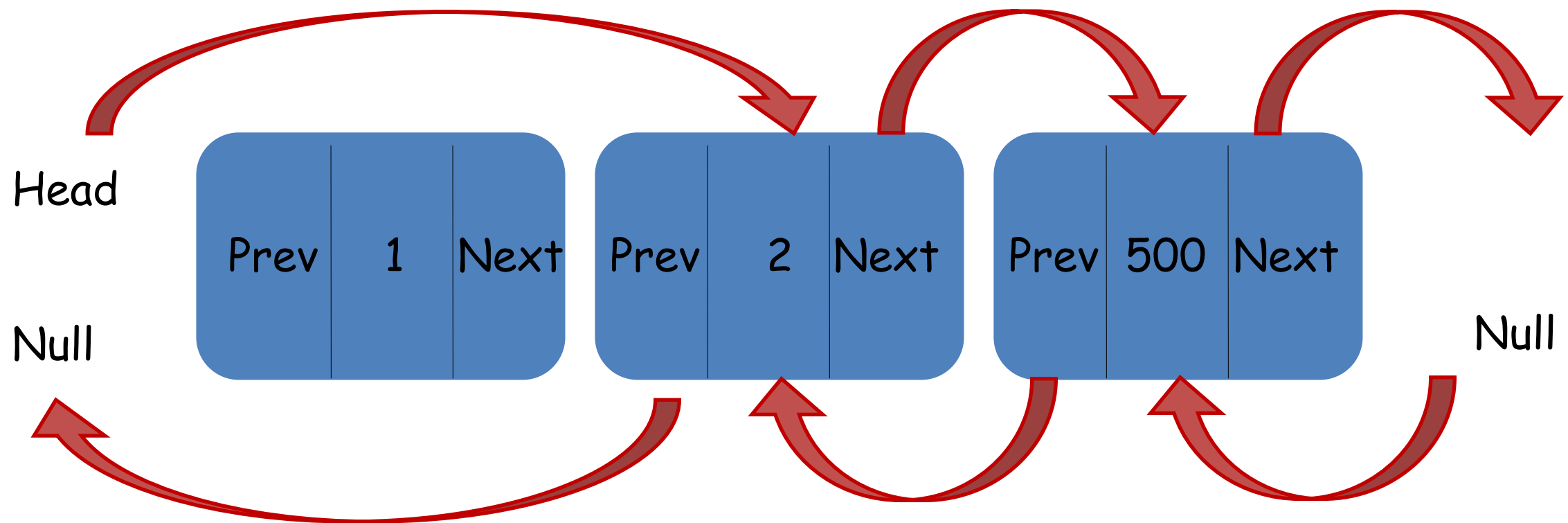
Doubley Linked List Data Structure



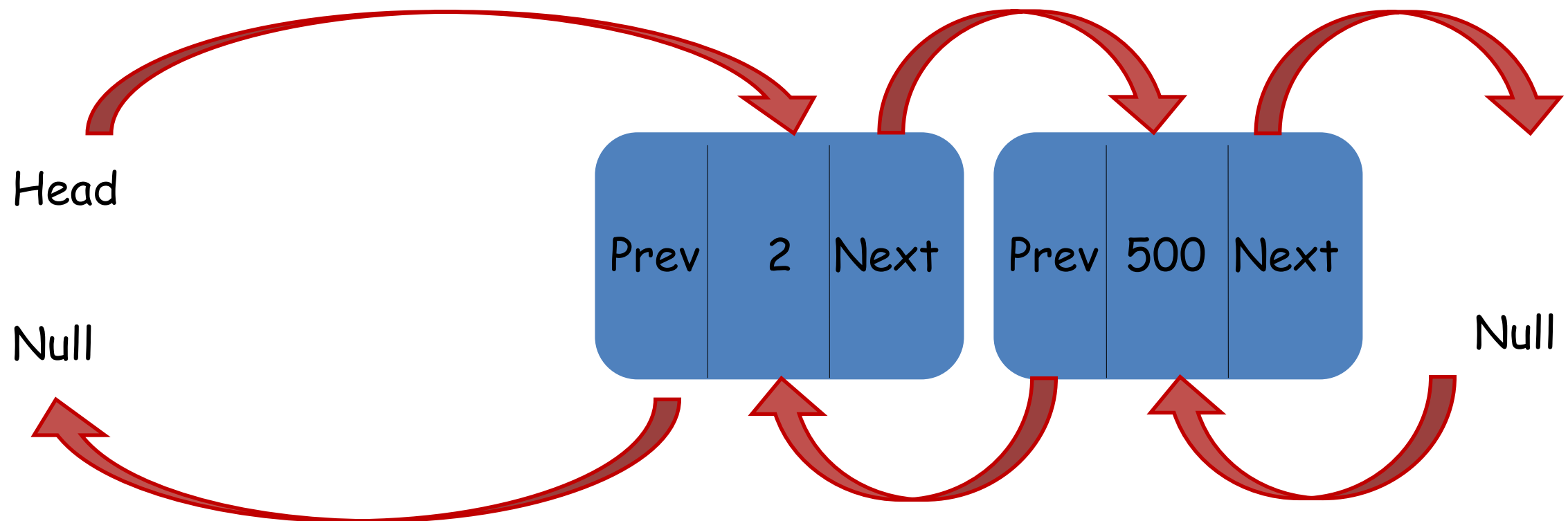
Doubley Linked List Data Structure



Doubley Linked List Data Structure



Doubley Linked List Data Structure



Definition

Removes the first node (head) from a doubly linked list and updates the new head correctly.

Explanation

- ◆ Deleting the first node requires updating:
 - ▣ `head` to point to the next node
 - ▣ `prev` of the new head to `NULL`
 - ▣ The old head must be deleted from memory

Procedures (Delete First Node)

1. Check if list is empty

 Check:

 If `head == NULL`

 → Result: Stop if the list is empty

2. Store the Old Head

 Step:

 Store `head` in a temporary pointer `temp`

 → Result: We keep a reference to the node that will be deleted

3. Move Head

 Step:

 Set `head = head->next`

 → Result: The second node becomes the new head

4. Update Previous Pointer

 If the new head exists:

 Set `head->prev = NULL`

 → Result: The new head has no previous node

5. Delete Old Head

 Step:

 `delete temp`

 → Result: The old first node is removed from memory

💡 Example

Before:

NULL <--> 1 <--> 2 <--> 3 <--> 4 <--> 5 <--> NULL

Delete the first node:

Delete 1

After:

NULL <--> 2 <--> 3 <--> 4 <--> 5 <--> NULL

Note

◆ `DeleteFirstNode` is different from `DeleteNode` because:

- ▣ It always removes the first node
- ▣ No `Find` operation is required
- ▣ No traversal is required



Comparison

▨ Delete First Node

- ✓ Time Complexity: $O(1)$
- ✓ No traversal needed
- ✓ Directly updates the head

▤ Delete Specific Node

- ✓ Time Complexity: $O(n)$ (if using Find)
- ✓ Requires finding the node first
- ✓ Updates surrounding pointers

⊘ Warning

- ⚠ Always check if `head == NULL` before deleting.
- ⚠ The new head's `prev` must be set to `NULL`.
- ⚠ Forgetting `delete temp` causes a memory leak.

✦✦ Summary

- ✓ DeleteFirstNode removes the first node from the list
- ✓ Updates head to the next node
- ✓ Sets the new head's prev to NULL
- ✓ Time Complexity is $O(1)$

Thank you 😊